Veronica Saz Ulibarrena

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Leiden, The Netherlands
Nationality : Spanish

I am an aerospace engineer and computational astrophysicist passionate about space exploration and complex problems. I have worked on interplanetary trajectory design, celestial mechanics, optimization, and machine learning. I want to grow in my career to lead a team of passionate people working on exciting, challenging projects that have a positive impact on the world.

Experience

Present – May 2021 | **Ph.D. researcher , Leiden Observatory, Leiden, The Netherlands**
Machine Learning for the Gravitational *N*-body problem.
- Understanding of the potential **applications** and **limitations** of Machine Learning tools such as Physics-informed Neural Networks and Reinforcement Learning for celestial dynamics.
- **Supervised** bachelor and master students.
- Developed communication skills by **presenting** my work at **international**, multi-disciplinary conferences such as Information Universe Conference (2021), NeurIPS (2022), ENUMATH (2023).
- Participated in international collaborations and **multi-disciplinary** teamwork.
- Contributed as part of the scientific **organization** committee for the AMUSE Summer School (2021).
- Software tools : `Python`, `Tensorflow`, `PyTorch`, `Git`

April 2020 – February 2020 | **Student Job | Python Modeller, Delft Offshore Turbine, Delft, The Netherlands**
- `Python` modeling of a system of offshore wind turbines.
- Learned about hydraulics, control systems, programming, and graphic interface development with `PyQt`.
- **Developed a product** (simulation and plotting code) based on the requirements of different team members.

November 2019 – August 2019 | **Master Internship, European Space Agency (ESA) at ESOC, Darmstadt, Germany**
Interplanetary trajectory optimization.
- **Tested and optimized** a range of selected trajectories for JUICE mission using ESA's newly developed software.
- Developed a **graphic user interface** for the optimization software.
- Learned about the stages of a **space mission** and worked with a team of international, interdisciplinary members.

Education

2023-2024 | **Advanced Project Management Course (12 h), Leiden Observatory, Leiden, The Netherlands**
Understanding and application of project management concepts to a Ph.D. research project.

2021 | **Summer School Science Communication (40 h), Leiden Observatory, Leiden, The Netherlands**
Theory of science communication, target audiences, and how to write a book, among others.

2018-2021 | **Master in Aerospace Engineering, Space Flight track, Space Exploration focus , Technical University of Delft, Delft, The Netherlands**
- Specialized in transfer orbits and optimization with courses such as Satellite Orbit Determination, Numerical Astrodynamics, Propagation and Optimization in Astrodynamics, Engineering Optimization, and Mission Geometry.
- Master thesis : application of Artificial Neural Networks to the **optimization** of interplanetary missions using low-thrust propulsion.
- Master internship at European Space Agency working on the optimization of the trajectory of JUICE.

2014-2018 | **Bachelor in Aerospace Engineering, University of Leon, Leon, Spain**
- Carried out independent study, developed cooperative projects, solved engineering-related problems, and attended aerospace-related lectures.
- Relevant courses : Flight mechanics, Aerodynamics, Jet Engines, Control Systems, **Modelling Complex Systems**.

2017-2018 | **Exchange year as part of the Bachelor's degree, University of Vermont, Vermont, USA**
Developed a practical approach to problems.

Technical Expertise

Text Editors	Latex, Word, Power Point
Programming	`Python`, `PyQt`, `Tensorflow`, `PyTorch`, `Matlab`, `C++`, `Pandas`, `Matplotlib`
Operating Systems	Ubuntu, Windows, Oracle Linux
Software tools	Excel, Inkscape, Visual Code Studio, Git, Github, Autocad, Simscale, Abaqus, Paraview

Languages

Spanish	●●●●●
English	●●●●○
Dutch	●●○○○
French	●○○○○

Key Strengths

- Communication skills
- Leadership, teamwork, and international experience
- Analytical skills
- Persistent, organized, independent, efficient

Projects

Coordinator of Astronomy on Tap Leiden — April 2022- Present
https://aot.strw.leidenuniv.nl/
Volunteering work on science divulgation for the general public. Public speaking and organization of speakers for monthly events.

Master Thesis, Delft University of Technology — June 2020 - April 2021
github.com/veronicasaz/MasterThesis TUDelft Repository
Low-Thrust Interplanetary Trajectory Optimization Using Pre-Trained Artificial Neural Network Surrogates
- Developed astrodynamics libraries for interplanetary trajectory design.
- Developed own optimization codes to adapt to problem requirements.
- Researched and applied machine learning techniques.

Bachelor Thesis, Vermont University - Universidad de Leon — September 2017 - July 2018
github.com/veronicasaz/https://github.com/veronicasaz/AstrodynamicsScripts
Application of Basin Hopping method to the optimization of fuel mass for the preliminary design of a low-thrust trajectory from Earth to Mars.
Developed astrodynamics libraries for interplanetary trajectory design. Carried out independent study and research.

National Competition of the AIAA (American Institute of Aeronautics and Astronautics) — September 2017 - July 2018
https://www.aiaa.org/
Member of the AIAA club at the University of Vermont. Participated in discussions, and shared aerospace-related news. Participated in the national competition : *Undergraduate Team Space Transportation - Pluto Orbiter*, cooperation and exchange of ideas with engineering students to create a possible mission to Pluto.

Organizer at AEPSA (Student Association for Promotion of Autonomous Systems) — September 2015 - June 2017
Student Association at the University of Leon.
Carried out quadcopter-assembly projects with other Aerospace Engineering students. Organized and taught courses for both adults and children related to UAV's (Unmanned Aerial Vehicles) legislation, safety, and technical characteristics.

Publications

January 2024 — A hybrid approach for solving the gravitational N-body problem with Artificial Neural Networks.
Journal of Computational Physics github.com/veronicasaz/PlanetarySystem_HNN

August 2023 — A Generalized Framework of Neural Networks for Hamiltonian Systems.
https://papers.ssrn.com

Interests and Activities

2001 - Present	Karate : Second Dan-black belt and regional-level competitions
Other	Traveling, sports, reading, photography, guitar playing

References

References available upon request